Postmortem: checkout outage on 12 March

Summary

For 41 minutes, about one in five checkout requests failed with a timeout. No orders were lost: clients retried, and every payment that went through was recorded once.

Timeline

14:02 a deploy doubled the number of order workers from 8 to 16.
14:09 the queue backlog alert fired; the oldest message was eleven minutes old.
14:20 on-call saw database connection errors in the worker logs.
14:43 the worker count was rolled back to 8 and the backlog drained in six minutes.

Root cause

Each worker opens a pool of 10 database connections. Sixteen workers asked for 160 connections, but the database allows 100. Workers that could not get a connection timed out, their messages went back to the queue, and the retries made the backlog grow.

What we changed

The worker pool size is now derived from the database connection limit divided by the worker count. The deploy pipeline checks that product before it scales workers. The runbook for the queue backlog alert now says to look at connection errors first.